SUPPORTING ADULT GUIDANCE

Ten times bigger, ten times smaller

Place Value · Lesson 2. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

In the main activity, the Sliding Digits machine, learners put a number into a place-value chart with a fixed decimal point, predict, then press × 10, × 100, × 1000, ÷ 10, ÷ 100 or ÷ 1000. Every digit slides the right number of columns, and grey zeros fade in only where a column has emptied. Seven questions follow, and a stretch explains where the zero in 360 comes from, ending with 36 ÷ 10 = 3.6 as a bridge to decimals. Learners then do quick calculations and missing-number problems, test the "add a zero" shortcut on cases where it works and fails, and reach targets with the machine's buttons.

Driving Question: *When I multiply by 10, what moves?*

The mathematics, in plain terms

Multiplying by 10■ shifts every digit n places to the left relative to the decimal point; dividing shifts them right. Zeros are placeholders for empty columns. "Add a zero" describes the result for whole numbers being multiplied, but it is not the mechanism, and it fails for decimals (2.5 × 10 ≠ 2.50) and for dividing numbers that do not end in enough zeros (4,050 ÷ 100 = 40.5). Decimals proper come in Lesson 4.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Multiplying by 10

What is 36 × 10, and where does the 0 come from?

36

Thousands	Hundreds	Tens	Ones
		3	6

36 × 10: every digit one column left

Thousands	Hundreds	Tens	Ones
	3	6	0

The highlighted 0 holds the ones column the 6 left empty.

1. 36 is	3 tens and 6 ones
2. × 10 moves each digit one column left.	3 hundreds and 6 tens
3. The ones column is now empty.	a 0 holds it
4. Read the new number.	360

36 × 10 = 360. The zero was not added: it holds the empty ones column.

Example 2 — Dividing by 100

What is 4,500 ÷ 100?

4,500

Thousands	Hundreds	Tens	Ones
4	5	0	0

4,500 ÷ 100: every digit two columns right

Thousands	Hundreds	Tens	Ones
		4	5

1. 4,500 is	4 thousands and 5 hundreds
2. ÷ 100 moves each digit two columns right.	4 tens and 5 ones
3. Check by undoing: × 100.	45 × 100 = 4,500

4,500 ÷ 100 = 45.

Watch it move, then explain it, then test the shortcut

Learners see the digits move before naming any rule, so the rule is something they observed. The stretch explains the placeholder zero. The myth-buster comes after, so learners can judge the shortcut against an idea they already trust.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Sliding Digits machine and seven questions. Stretch: where does the zero come from?	~16 min
Movement	Hop the columns: quick calculations and missing numbers.	~7 min
Reflection	"Add a zero?": test the shortcut.	~7 min
Creativity	Reach the target with the machine's buttons; find the single move.	~10 min
Rest	A number can grow a thousand times without a digit changing.	~2 min
Nutrition	Intellectual: understanding over shortcuts.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Adding a zero to the end for any × 10, including decimals (2.5 × 10 = 2.50).
Ask: “Is 2.50 bigger than 2.5? Where should each digit go?”

Moving the digits the wrong way (× 10 giving a smaller number).
Ask: “Should multiplying by 10 make the number bigger or smaller?”

Moving one column for × 100.
Ask: “How many tens make a hundred? So how many columns?”

Removing any zero when dividing (4,050 ÷ 100 = 45).
Ask: “Which columns are those zeros in? What happens to the 5?”

Thinking the decimal point moves.
Ask: “Watch the point on the machine. Does it move, or do the digits?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: what each column is worth (Lesson 1). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner writes 47 × 100 = 470. The aim is a question that lets them see the second column.

- “How many columns did the digits move?”
- “How many columns does × 10 move them? And × 100?”

- “Can you try it on the machine and watch?”
- “How could you check your answer by dividing?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Should learners say "the decimal point moves"?

No. The point stays still and the digits move; that is the model learners will need for decimals and standard form.

Why include decimals at all in this lesson?

Only as a glimpse: 36 ÷ 10 = 3.6 in the stretch, and one practice question. Lesson 4 teaches decimals properly.

Is the stretch required?

No. The secure core is multiplying and dividing whole numbers by 10, 100 and 1000 and explaining which way the digits move.

What comes next?

Lesson 3: zero as a placeholder.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.

Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

NEO by Nudge Education · nudgeeducation.online · Curriculum by Gerry Docherty. Supporting Adult Guidance v0.1.